5.3.1 Transition elements

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
Properties		
(a)	the electron configuration of atoms and ions of the d-block elements of Period 4 (Sc–Zn), given the atomic number and charge **(see also 2.2.1 d)**	Learners should use sub-shell notation e.g. for Fe: $1s^2 2s^2 2p^6 3s^2 3p^6 3d^6 4s^2$.
(b)	the elements Ti–Cu as transition elements i.e. d-block elements that have an ion with an incomplete d-sub-shell	
(c)	illustration, using at least two transition elements, of: **(i)** the existence of more than one oxidation state for each element in its compounds **(see also 5.3.1 k)** (ii) the formation of coloured ions **(see also 5.3.1 h, j–k)** (iii) the catalytic behaviour of the elements and their compounds and their importance in the manufacture of chemicals by industry **(see 3.2.2 d)**	No detail of how colour arises required. Practical examples of catalytic behaviour include: Cu^{2+} for reaction of Zn with acids; MnO_2 for decomposition of H_2O_2. No detail of catalytic processes required. HSW9 Benefits of reduced energy usage; risks from toxicity of many transition metals.

Ligands and complex ions

(d)	explanation and use of the term *ligand* in terms of coordinate (dative covalent) bonding to a metal ion or metal, including bidentate ligands	Examples should include: monodentate: H_2O, Cl^- and NH_3 bidentate: $NH_2CH_2CH_2NH_2$ ('en'). In exams, other ligands could be introduced.
(e)	use of the terms *complex ion* and *coordination number* and examples of complexes with: **(i)** six-fold coordination with an octahedral shape (ii) four-fold coordination with either a planar or tetrahedral shape **(see also 2.2.2 g–h)**	*M4.1, M4.2* Examples: Octahedral: many hexaaqua complexes, e.g. $[Cu(H_2O)_6]^{2+}$, $[Fe(H_2O)_6]^{3+}$ Tetrahedral: many tetrachloro complexes, e.g. $CuCl_4^{2-}$ and $CoCl_4^{2-}$ Square planar: complexes of Pt, e.g. platin: $Pt(NH_3)_2Cl_2$ **(see also 5.3.1 g)**.
(f)	types of stereoisomerism shown by complexes, including those associated with bidentate and multidentate ligands: **(i)** *cis–trans* isomerism e.g. $Pt(NH_3)_2Cl_2$ **(see also 4.1.3 c–d)** (ii) optical isomerism e.g. $[Ni(NH_2CH_2CH_2NH_2)_3]^{2+}$ **(see also 6.2.2 c)**	*M4.1, M4.2, M4.3* Learners should be able to draw 3-D diagrams to illustrate stereoisomerism. HSW8
(g)	use of *cis*-platin as an anti-cancer drug and its action by binding to DNA preventing cell division	HSW9 Benefits of chemotherapy; risks from unpleasant side effects.

Ligand substitution

(h)	ligand substitution reactions and the accompanying colour changes in the formation of: **(i)** $[Cu(NH_3)_4(H_2O)_2]^{2+}$ and $[CuCl_4]^{2-}$ from $[Cu(H_2O)_6]^{2+}$ (ii) $[Cr(NH_3)_6]^{3+}$ from $[Cr(H_2O)_6]^{3+}$ **(see also 5.3.1 j)**	Complexed formulae should be used in ligand substitution equations.
(i)	explanation of the biochemical importance of iron in haemoglobin, including ligand substitution involving O_2 and CO	

Precipitation reactions

(j)	reactions, including ionic equations, and the accompanying colour changes of aqueous Cu^{2+}, Fe^{2+}, Fe^{3+}, Mn^{2+} and Cr^{3+} with aqueous sodium hydroxide and aqueous ammonia, including: **(i)** precipitation reactions (ii) complex formation with excess aqueous sodium hydroxide and aqueous ammonia	For precipitation, non-complexed formulae or complexed formulae, are acceptable e.g. Cu^{2+}(aq) or $[Cu(H_2O)_6]^{2+}$; $Cu(OH)_2$(s) or $Cu(OH)_2(H_2O)_4$. With excess NaOH, only $Cr(OH)_3$ reacts further forming $[Cr(OH)_6]^{3-}$. With excess NH_3, only $Cr(OH)_3$ and $Cu(OH)_2$ react forming $[Cr(NH_3)_6]^{3+}$ and $[Cu(NH_3)_4(H_2O)_2]^{2+}$ respectively **(see also 5.3.1 h)**.

Redox reactions

(k) redox reactions and accompanying colour changes for:

(i) interconversions between Fe^{2+} and Fe^{3+}

(ii) interconversions between Cr^{3+} and $Cr_2O_7^{2-}$

(iii) reduction of Cu^{2+} to Cu^+ and disproportionation of Cu^+ to Cu^{2+} and Cu

Fe^{2+} can be oxidised with H^+/MnO_4^- and Fe^{3+} reduced with I^-, Cr^{3+} can be oxidised with H_2O_2/OH^- and $Cr_2O_7^{2-}$ reduced with Zn/H^+, Cu^{2+} can be reduced with I^-. In aqueous conditions, Cu^+ readily disproportionates.

Learners will **not** be required to recall equations but may be required to construct and interpret redox equations using relevant half-equations and oxidation numbers **(see 5.2.3 b–c)**.

(l) interpretation and prediction of unfamiliar reactions including ligand substitution, precipitation, redox.